

X_1	X_2	X_3	$X_1 X_2$	$X_1 X_3$	$X_2 X_3$	X_1^2	X_2^2	X_3^2

$$\bar{X}_1 = \frac{\sum X_1}{n} \quad S_1^2 = \frac{\sum X_1^2}{n} - \left(\frac{\sum X_1}{n}\right)^2$$

$$S_2^2 = \frac{\sum X_2^2}{n} - \left(\frac{\sum X_2}{n}\right)^2$$

$$\bar{X}_2 = \frac{\sum X_2}{n}$$

$$\bar{X}_3 = \frac{\sum X_3}{n}$$

$$S_3^2 = \frac{\sum X_3^2}{n} - \left(\frac{\sum X_3}{n}\right)^2$$

Now:

$$\delta_{12} = \frac{n \sum X_1 X_2 - \sum X_1 \cdot \sum X_2}{\sqrt{[n \sum X_1^2 - (\sum X_1)^2][n \sum X_2^2 - (\sum X_2)^2]}}$$

$$\delta_{13} = \frac{n \sum X_1 X_3 - \sum X_1 \cdot \sum X_3}{\sqrt{[n \sum X_1^2 - (\sum X_1)^2][n \sum X_3^2 - (\sum X_3)^2]}}$$

$$\delta_{23} = \frac{n \sum X_2 X_3 - \sum X_2 \cdot \sum X_3}{\sqrt{[n \sum X_2^2 - (\sum X_2)^2][n \sum X_3^2 - (\sum X_3)^2]}}$$

Regression equation of X_1 on X_2 and X_3

$$\frac{X_1 - \bar{X}_1}{S_1} = \left(\frac{\delta_{12} - \delta_{13} \delta_{23}}{1 - \delta_{23}^2} \right) \left(\frac{X_2 - \bar{X}_2}{S_2} \right) + \left(\frac{\delta_{13} - \delta_{12} \delta_{23}}{1 - \delta_{23}^2} \right) \left(\frac{X_3 - \bar{X}_3}{S_3} \right)$$

Regression equation of X_2 on X_1 and X_3

$$\frac{X_2 - \bar{X}_2}{S_2} = \left(\frac{\delta_{12} - \delta_{13} \delta_{23}}{1 - \delta_{13}^2} \right) \left(\frac{X_1 - \bar{X}_1}{S_1} \right) + \left(\frac{\delta_{23} - \delta_{12} \delta_{13}}{1 - \delta_{13}^2} \right) \left(\frac{X_3 - \bar{X}_3}{S_3} \right)$$

Regression eq. of X_3 on X_1 and X_2 .

$$\frac{X_3 - \bar{X}_3}{S_3} = \left(\frac{\delta_{13} - \delta_{12} \delta_{23}}{\delta_{12}^2} \right) \left(\frac{X_1 - \bar{X}_1}{S_1} \right) + \left(\frac{\delta_{23} - \delta_{12} \delta_{13}}{1 - \delta_{12}^2} \right) \left(\frac{X_2 - \bar{X}_2}{S_2} \right)$$

Multiple correlation coefficient.

$$R_{1.23} = \sqrt{\frac{\sigma_{12}^2 + \sigma_{13}^2 - 2\sigma_{12}\sigma_{23}\sigma_{13}}{1 - \sigma_{23}^2}}$$

11.12 ③ all part

$$R_{2.13} = \sqrt{\frac{\sigma_{12}^2 + \sigma_{23}^2 - 2\sigma_{12}\sigma_{23}\sigma_{13}}{1 - \sigma_{13}^2}}$$

$$R_{3.12} = \sqrt{\frac{\sigma_{13}^2 + \sigma_{23}^2 - 2\sigma_{12}\sigma_{23}\sigma_{13}}{1 - \sigma_{12}^2}}$$

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Partial correlation coefficient

$$\sigma_{12.3} = \frac{\sigma_{12} - \sigma_{13}\sigma_{23}}{\sqrt{(1 - \sigma_{13}^2)(1 - \sigma_{23}^2)}}$$

$$\sigma_{13.2} = \frac{\sigma_{13} - \sigma_{12}\sigma_{23}}{\sqrt{(1 - \sigma_{12}^2)(1 - \sigma_{23}^2)}}$$

$$\sigma_{23.1} = \frac{\sigma_{23} - \sigma_{12}\sigma_{13}}{\sqrt{(1 - \sigma_{12}^2)(1 - \sigma_{13}^2)}}$$

If Regression is (-ve) then correlation can never be positive.

$$b_{12} = -0.20$$

$$b_{21} = -0.71$$

$$\sigma_{12} = \pm \sqrt{b_{12} \times b_{21}} \\ = \sqrt{(-0.20)(-0.70)}$$

Example 11.4

$$\bar{X}_1 = 6.8 \quad S_1 = 1.0 \quad \sigma_{12} = 0.60$$

$$\bar{X}_2 = 7.0 \quad S_2 = 0.8 \quad \sigma_{13} = 0.70$$

$$\bar{X}_3 = 7.4 \quad S_3 = 0.9 \quad \sigma_{23} = 0.65$$

a) Find least square regression equ. of X_2 on X_1 and X_3

b) Estimates to find grades of two students who scored (1) 9 and 7, and (2) 4 and 8 on two quizzes

c) Compute $R_{2.13}$
3.12

a) Solution: The regression equ. of X_3 on X_1 and X_2 are.

$$\frac{X_3 - \bar{X}_3}{S_3} = \left(\frac{\delta_{13} - \delta_{12}\delta_{23}}{1 - \delta_{12}^2} \right) \left(\frac{X_1 - \bar{X}_1}{S_1} \right) + \left(\frac{\delta_{23} - \delta_{12}\delta_{13}}{1 - \delta_{12}^2} \right) \left(\frac{X_2 - \bar{X}_2}{S_2} \right)$$

~~$X_3 = 74$~~ first we calculate.

$$\frac{\delta_{13} - \delta_{12}\delta_{23}}{1 - \delta_{12}^2}$$

$$\frac{X_3 - 74}{9.0} = \frac{0.70 - (0.60)(0.65)}{1 - (0.60)^2} \left(\frac{X_1 - 6.8}{1.8} \right) + \frac{0.65 - (0.60)(0.70)}{1 - (0.60)^2} \left(\frac{X_2 - 7.0}{0.8} \right)$$

$$X_3 - 74 = 4.36 (X_1 - 6.8) + 4.04 (X_2 - 7.0)$$

$$X_3 - 74 = 4.36 (X_1 - 6.8) + 4.04 (X_2 - 7.0)$$

$$X_3 - 74 = 4.36 X_1 - 29.65 + 4.04 X_2 - 28.28$$

$$X_3 = 16.07 + 4.36 X_1 + 4.04 X_2$$

b) Student 1: when $X_1 = 9$ and $X_2 = 7$ we get.

$$\hat{X}_3 = 16.07 + 39.24 + 28.28 \Rightarrow \hat{X}_3 = 84$$

Student 2: when $X_1 = 4$ and $X_2 = 8$

$$\hat{X}_3 = 66$$

$$g) R_{3.12} = \frac{\delta_{13}^2 + \delta_{23}^2 - 2\delta_{12}\delta_{13}\delta_{23}}{1 - \delta_{12}^2} = \frac{(0.70)^2 + (0.65)^2 - 2(0.60)(0.70)(0.65)}{1 - (0.60)^2}$$

$$= 0.7567$$

X_1	X_2	X_3	$X_1 X_2$	$X_1 X_3$	$X_2 X_3$	X_1^2	X_2^2	X_3^2
3	16	90	48			9		
5	10	72	50			25		
6	7	54	42			36		
8	4	42	32			64		
12	3	30	36			144		
14	2	12	28			196		
48	42	300	236	1818	2820	474	434	1908

$$\bar{X}_1 = 8$$

$$S_1 = 3.87$$

Calculator (C)

$$\bar{X}_2 = 7$$

$$S_2 = 4.83$$

$$\bar{X}_3 = 50$$

$$S_3 = 25.85$$

$$r_{12} = \frac{n \sum x_1 x_2 - \sum x_1 \cdot \sum x_2}{\sqrt{[n \sum x_1^2 - (\sum x_1)^2]} \sqrt{[n \sum x_2^2 - (\sum x_2)^2]}} = \frac{-600}{6515 \times 25210}$$



11.8 (a)

X_1	X_2	X_3	$X_1 X_2$	$X_1 X_3$	$X_2 X_3$	X_1^2	X_2^2	X_3^2
52	3	2						
18	2	4						
52	5	2						
16	1	5						
42	4	3						
42	6	9						
208	21	25	864	878	95	8376	91	139

$$\bar{X}_1 = 34.67 \quad \bar{X}_2 = 3.5 \quad \bar{X}_3 = 4.17$$

$$r_{12} = \frac{n \sum X_1 X_2 - \sum X_1 \cdot \sum X_2}{\sqrt{[n \sum X_1^2 - (\sum X_1)^2][n \sum X_2^2 - (\sum X_2)^2]}} = \frac{6(864) - (208)(21)}{\sqrt{[6(8376) - (208)^2][6(91) - (21)^2]}} = 0.95$$

$$r_{13} = \frac{n \sum X_1 X_3 - \sum X_1 \cdot \sum X_3}{\sqrt{[n \sum X_1^2 - (\sum X_1)^2][n \sum X_3^2 - (\sum X_3)^2]}} = \frac{6(878) - (208)(25)}{\sqrt{[6(8376) - (208)^2][6(139) - (25)^2]}}$$

$$r_{13} = 0.056$$

$$r_{23} = \frac{n \sum X_2 X_3 - \sum X_2 \cdot \sum X_3}{\sqrt{[n \sum X_2^2 - (\sum X_2)^2][n \sum X_3^2 - (\sum X_3)^2]}} = \frac{6(95) - (21)(25)}{\sqrt{[6(91) - (21)^2][6(139) - (25)^2]}} = 0.30$$

11.8 (b)

$$\bar{X}_1 = 20, \quad \bar{X}_2 = 36, \quad \bar{X}_3 = 12$$

$$S_1 = 1.0, \quad S_2 = 2.0, \quad S_3 = 1.5$$

$$r_{12} = -0.20, \quad r_{13} = 0.40, \quad r_{23} = 0.50$$

Find regression eq. X_3 on X_1 and X_2 .

$$\frac{X_3 - \bar{X}_3}{S_3} = \left(\frac{r_{13} - r_{12} r_{23}}{(1 - r_{12}^2)} \right) \left(\frac{X_1 - \bar{X}_1}{S_1} \right) + \left(\frac{r_{23} - r_{12} r_{13}}{(1 - r_{12}^2)} \right) \left(\frac{X_2 - \bar{X}_2}{S_2} \right)$$

$$\frac{r_{13} - r_{12} r_{23}}{(1 - r_{12}^2)} = \frac{0.40 - (-0.20)(0.50)}{1 - (-0.20)^2} = \frac{0.5}{0.96}$$

$$\frac{r_{23} - r_{12} r_{13}}{(1 - r_{12}^2)} = \frac{0.50 - (-0.20)(0.40)}{1 - (-0.20)^2} = \frac{0.58}{0.96}$$

$$\frac{X_3 - 12}{1.5} = \frac{0.5}{0.96} \left(\frac{X_1 - 20}{1.0} \right) + \frac{0.58}{0.96} \left(\frac{X_2 - 36}{2.0} \right)$$

$$X_3 - 12 = 0.78125(X_1 - 20) + 0.453125(X_2 - 36)$$

$$= 0.78125X_1 - 15.625 + 0.453125X_2 - 16.3125$$

$$X_3 = -19.94 + 0.78X_1 + 0.45X_2$$

(Q# 11.12)

Find the Anceq. on X_1 on X_2 and X_3 .

$$\frac{X_1 - \bar{X}_1}{S_1} = \left(\frac{\delta_{12} - \delta_{13}\delta_{23}}{1 - \delta_{23}^2} \right) \left(\frac{X_2 - \bar{X}_2}{S_2} \right) + \left(\frac{\delta_{13} - \delta_{12}\delta_{23}}{S_2} \right) \left(\frac{X_3 - \bar{X}_3}{S_3} \right)$$

$$\bar{X}_1 = 28.02, \quad \bar{X}_2 = 4.91, \quad \bar{X}_3 = 594, \quad S_1 = 4.42, \quad S_2 = 1.10, \quad S_3 = 85$$

$$\delta_{12} = 0.80, \quad \delta_{13} = -0.40, \quad \delta_{23} = -0.56$$

$$\frac{X_1 - 28.02}{4.42} = \left(\frac{0.80 - (-0.40)(-0.56)}{1 - (-0.56)^2} \right) \left(\frac{X_2 - 4.91}{1.10} \right) + \left(\frac{-0.40 - (0.80)(-0.56)}{1 - (-0.56)^2} \right) \left(\frac{X_3 - 594}{85} \right)$$

$$= \left(\frac{120}{143} \right) \left(\frac{X_2 - 4.91}{1.10} \right) + \left(\frac{10}{143} \right) \left(\frac{X_3 - 594}{85} \right)$$

$$= 3.37(X_2 - 4.91) + 0.0036(X_3 - 594)$$

$$X_1 - 28.02 = 3.37X_2 - 16.51 + 0.0036X_3 - 2.14$$

$$X_1 = 3.37X_2 + 0.0036X_3 - 18.65$$

$$X_1 = 9.37 + 3.37X_2 + 0.0036X_3$$

Partial correlation co-eff.

$$\delta_{12.3} = \frac{\delta_{12} - \delta_{13}\delta_{23}}{\sqrt{(1 - \delta_{13}^2)(1 - \delta_{23}^2)}}, \quad \delta_{13.2} = \frac{\delta_{13} - \delta_{12}\delta_{23}}{\sqrt{(1 - \delta_{12}^2)(1 - \delta_{23}^2)}}, \quad \delta_{23.1} = \frac{\delta_{23} - \delta_{12}\delta_{13}}{\sqrt{(1 - \delta_{12}^2)(1 - \delta_{13}^2)}}$$

$$\text{now } \delta_{12} = 0.80, \quad \delta_{13} = -0.40, \quad \delta_{23} = -0.56$$

$$\delta_{12.3} = \frac{\delta_{12} - \delta_{13}\delta_{23}}{\sqrt{(1 - \delta_{13}^2)(1 - \delta_{23}^2)}} \Rightarrow \delta_{12.3} = \frac{(0.80) - (-0.40)(-0.56)}{\sqrt{1 - (-0.40)^2} \sqrt{1 - (-0.56)^2}} = 0.80$$

$$\delta_{13.2} = \frac{\delta_{13} - \delta_{12}\delta_{23}}{\sqrt{(1 - \delta_{12}^2)(1 - \delta_{23}^2)}} = \frac{-0.40 - (0.80)(-0.56)}{\sqrt{1 - (0.80)^2} \sqrt{1 - (-0.56)^2}} = 0.047$$

$$\delta_{23.1} = \frac{\delta_{23} - \delta_{12}\delta_{13}}{\sqrt{(1 - \delta_{12}^2)(1 - \delta_{13}^2)}} = \frac{-0.56 - (0.80)(-0.40)}{\sqrt{1 - (0.80)^2} \sqrt{1 - (-0.40)^2}} = -0.436$$

11.9(b)

$$b_{21} = 0.75 \quad b_{22} = 0.88 \quad b_{13} = 0.58 \quad b_{31} = 1.88$$

$$b_{23} = 0.53 \quad b_{32} = 1.30$$

Multiple correlation of X_2 with X_1 and X_3

$$r_{12} = \pm \sqrt{b_{12} \times b_{21}}$$

$$r_{12} = \pm \sqrt{0.75 \times 0.88}$$

$$\Rightarrow r_{12} = 0.81$$

$$r_{13} = \pm \sqrt{b_{13} \times b_{31}}$$

$$r_{13} = \pm \sqrt{0.58 \times 1.68}$$

$$\Rightarrow r_{13} = 0.99$$

$$r_{23} = \pm \sqrt{b_{23} \times b_{32}}$$

$$= \sqrt{0.53 \times 1.30}$$

$$\Rightarrow r_{23} = 0.83$$

$$R_{2.13} = \frac{r_{12}^2 + r_{23}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{13}^2}$$

$$= \frac{(0.81)^2 + (0.83)^2 - 2(0.81)(0.83)(0.99)}{1 - (0.99)^2}$$

$$R_{2.13} = 0.834$$

(c) $r_{12} = 0.60, \quad r_{13} = 0.70, \quad r_{23} = 0.65$

$$R_{2.13} = \frac{r_{12}^2 + r_{23}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{13}^2}$$

$$= \frac{(0.60)^2 + (0.65)^2 - 2(0.60)(0.70)(0.65)}{1 - (0.70)^2}$$

$$R_{2.13} = 0.68$$

(11.10)

$$r_{12} = 0.8, \quad r_{13} = -0.7, \quad r_{23} = -0.9$$

MC-coe or $R_{1.23}$

$$R_{1.23} = \frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}$$

$$= \frac{0.8^2 + (-0.7)^2 - 2(0.8)(-0.7)(-0.9)}{1 - (-0.9)^2}$$

$$R_{1.23} = 0.80$$

b) MC-coeff $R_{2.13}$ and p.c-coef $r_{23.1}$ (X_2 on X_1 and X_3)

$$b_{12} = -0.1, \quad b_{21} = -0.4, \quad b_{13} = 0.27, \quad b_{31} = 0.6, \quad b_{23} = 0.67, \quad b_{32} = 0.38$$

$$r_{12} = \pm \sqrt{b_{12} \times b_{21}}$$

$$\Rightarrow r_{12} = -\sqrt{-0.1 \times -0.4} \Rightarrow r_{12} = -0.2$$

$$r_{13} = \pm \sqrt{b_{13} \times b_{31}}$$

$$= \sqrt{0.27 \times 0.6}$$

$$\Rightarrow r_{13} = 0.40$$

$$r_{23} = \pm \sqrt{b_{23} \times b_{32}}$$

$$= \sqrt{0.67 \times 0.38}$$

$$\Rightarrow r_{23} = 0.50$$

$$R_{2.13} = \frac{r_{12}^2 + r_{23}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{13}^2}$$

$$= \frac{(-0.2)^2 + (0.50)^2 - 2(-0.2)(0.40)(0.50)}{1 - (0.40)^2}$$

$$R_{2.13} = 0.66$$

Partial

$$r_{23.1} = \frac{r_{23} - r_{12}r_{13}}{\sqrt{[1-r_{12}^2][1-r_{13}^2]}} = \frac{0.5 - (0.40)(-0.20)}{\sqrt{[1-(0.40)^2][1-(0.40)^2]}} = 0.65$$

$$= (11.13) \rightarrow \star$$

$$\bar{X}_1 = 18.5 \quad S_1 = 11.2 \quad r_{12} = 0.60 \quad X_1 \text{ on } X_2 \text{ and } X_3$$

$$\bar{X}_2 = 100.6 \quad S_2 = 15.8 \quad r_{13} = 0.32 \quad X_2 \text{ on } X_1 \text{ and } X_3$$

$$\bar{X}_3 = 24 \quad S_3 = 6.0 \quad r_{23} = 0.35 \quad X_3 \text{ on } X_1 \text{ and } X_2$$

$$\frac{X_1 - \bar{X}_1}{S_1} = \left(\frac{r_{12} - r_{13}r_{23}}{1 - r_{23}^2} \right) \left(\frac{X_2 - \bar{X}_2}{S_2} \right) + \left(\frac{r_{13} - r_{12}r_{23}}{1 - r_{23}^2} \right) \left(\frac{X_3 - \bar{X}_3}{S_3} \right)$$

2- first pos bracket
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$$\frac{X_1 - 18.5}{11.2} = \left(\frac{0.60 - (0.32)(0.35)}{1 - (0.35)^2} \right) \left(\frac{X_2 - 100.6}{15.8} \right) + \left(\frac{0.32 - (0.60)(0.35)}{1 - (0.35)^2} \right) \left(\frac{X_3 - 24}{6.0} \right)$$

$$\textcircled{2} \quad = 11.2 \left(\frac{0.60 - (0.32)(0.35)}{1 - (0.35)^2} \right) \left(\frac{X_2 - 100.6}{15.8} \right) + 11.2 \left(\frac{0.32 - (0.60)(0.35)}{1 - (0.35)^2} \right) \left(\frac{X_3 - 24}{6.0} \right)$$

$$= 0.39(X_2 - 100.6) + 0.2339(X_3 - 24)$$

$$X_1 - 18.5 = 0.39X_2 - 39.23 + 0.2339X_3 - 5.61$$

$$X_1 = 0.39X_2 + 0.2339X_3 - 26.34$$

$$X_1 = -26.34 + 0.39X_2 + 0.23X_3$$

$$\frac{X_2 - \bar{X}_2}{S_2} = \left(\frac{r_{12} - r_{13}r_{23}}{1 - r_{13}^2} \right) \left(\frac{X_1 - \bar{X}_1}{S_1} \right) + \left(\frac{r_{23} - r_{12}r_{13}}{1 - r_{13}^2} \right) \left(\frac{X_3 - \bar{X}_3}{S_3} \right)$$

$$\frac{X_2 - 100.6}{15.8} = \left(\frac{0.60 - (0.32)(0.35)}{1 - (0.32)^2} \right) \left(\frac{X_1 - 18.5}{11.2} \right) + \left(\frac{0.35 - (0.32)(0.60)}{1 - (0.32)^2} \right) \left(\frac{X_3 - 24}{6.0} \right)$$

$$X_2 - 100.6 = 0.77(X_1 - 18.5) + 0.46(X_3 - 24)$$

$$= 0.77X_1 - 14.25 + 0.46X_3 - 11.04$$

$$X_2 = 75.34 + 0.77X_1 + 0.46X_3$$

$$\textcircled{1} \quad R_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{[1-r_{13}^2][1-r_{23}^2]}} =$$

$$R_{13} \quad R_{12.3} = 0.55$$

$$R_{13.2} = \frac{r_{13} - r_{12}r_{23}}{\sqrt{[1-r_{12}^2][1-r_{23}^2]}} =$$

$$R_{13.2} = 0.15$$

$$R_{23.1} = \frac{r_{23} - r_{12}r_{13}}{\sqrt{[1-r_{12}^2][1-r_{13}^2]}} =$$

$$R_{23.1} = 0.20$$

Testing hypothesis about Multiple correlation Co-efficient.

In population is zero.

(1) Formulation of hypothesis:

$$H_0: \rho_{y.123\dots p} = 0$$

$$H_1: \rho_{y.123\dots p} \neq 0$$

(2) Level of significance:

$$\alpha = 0.05$$

(3) F Test statistics:

$$F = \frac{R^2_{y.12\dots p} (n-p-1)}{(1-R^2_{y.12\dots p}) p}$$

(4) Critical region:

$$F \geq F_{\alpha}(\nu_1, \nu_2)$$

$$\nu_1 = p$$

$$\nu_2 = n-p-1$$

(5) Conclusion calculation:

(6) Conclusion:

$$21.31$$

$$n=20,$$

$$R_{1.23} = 0.35$$

(1) Formulation of hypothesis:

$$H_0: \rho_{1.23} = 0$$

$$H_1: \rho_{1.23} \neq 0$$

(2) L.O.S : $\alpha = 0.05$

(3) Test statistics:

$$F = \frac{R^2_{1.23} (n-p-1)}{(1-R^2_{1.23}) p}$$

(4) Critical region:

$$F \geq F_{\alpha}(\nu_1, \nu_2)$$

$$F \geq F_{0.05}(2, 17)$$

$$F \geq 3.59$$

(5) Calculation:

$$F = \frac{(0.35)^2 (20-2-1)}{(1-0.35)^2 (2)} = 2.46$$

(6) Conclusion:

Since the computed value $f = 2.46$ doesn't fall in C.R so we can accept H_0 . No significant diff.

General procedure of testing hypothesis that

Partial correlation co-efficient of order k in the population is zero.

① Formulation of hypothesis:

$$H_0: \rho_{12.345\dots P} = 0$$

$$H_1: \rho_{12.345\dots P} \neq 0$$

② Level of significance:

$$\alpha = 0.05$$

③ Test statistics:

$$t = \frac{r_{12.345\dots P} \sqrt{n-k-2}}{\sqrt{1-r_{12.345\dots P}^2}}$$

$$F = \frac{r_{12.345\dots P}^2 (n-k-2)}{1-r_{12.345\dots P}^2}$$

$$F \geq F_{\alpha}(v_1, v_2)$$

$$v_1 = 1$$

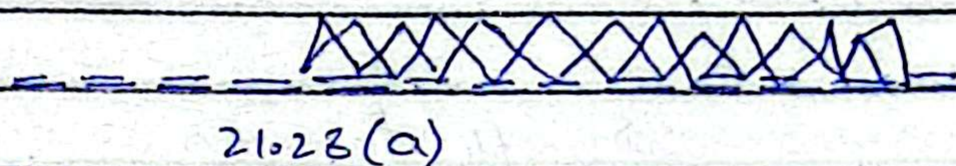
$$v_2 = n-k-2$$

④ Critical region:

$$|t| \geq t_{\alpha/2}(n-k-2)$$

⑤ Calculation.

⑥ Conclusion.



Given that: $r_{12.34} = 0.5$, $n=20$, $\alpha=5\%$

① F.o. H_1 :

$H_0: \rho_{12.34} = 0$ (not significant).

$H_1: \rho_{12.34} \neq 0$ (significant).

② L.O.S:

$$\alpha = 0.05$$

③ T.S:

$$t = \frac{r_{12.34} \sqrt{n-k-2}}{\sqrt{1-r_{12.34}^2}}$$

④ C.R:

$$|t| \geq t_{\alpha/2}(n-k-2)$$

$$|t| \geq t_{0.025}(20-2-2) \Rightarrow |t| \geq t_{0.025}(16)$$

$$|t| \geq 2.12$$

⑤ Calculation:

$$t = \frac{0.5 \times \sqrt{16}}{\sqrt{1-(0.5)^2}} = 2.31$$

⑥ Since the computed value of $t=2.31$ falls in C.R so we can't accept H_0 therefore we reject H_0 .

b₁₁

$$n = 25, \quad \delta_{12.34} = 0.48, \quad \alpha = 0.05$$

1) F.O.H:

$$H_0: \beta_{12.34} = 0$$

$$H_1: \beta_{12.34} \neq 0$$

2) L.O.S:

$$\alpha = 0.05$$

3) T.S:

$$t = \frac{\beta_{12.34} \sqrt{n-k-2}}{\sqrt{1 - \delta_{12.34}^2}}$$

4) C.R:

$$|t| \geq t_{\alpha/2}(n-k-2)$$

$$\Rightarrow |t| \geq t_{0.025}(25-2-2)$$

$$|t| \geq t_{0.025}(19) \Rightarrow |t| \geq 2.09$$

5) Calculation:

$$t = \frac{0.48 \times \sqrt{19}}{\sqrt{1 - (0.48)^2}} \Rightarrow t = 2.38$$

6) Conclusion: Since the computed value $t = 2.38$ falls in C.R
So we can reject H_0 .

21.29

$$\text{Given that: } n = 20, \quad \delta_{12.34} = 0.51$$

1) F.O.H:

$$H_0: \beta_{12.34} = 0$$

$$H_1: \beta_{12.34} \neq 0$$

2) L.O.S:

$$\alpha = 0.05$$

3) T.S:

$$t = \frac{\delta_{12.34} \sqrt{n-k-2}}{\sqrt{1 - \delta_{12.34}^2}}$$

4) C.R:

$$|t| \geq t_{\alpha/2}(n-2-k)$$

$$|t| \geq t_{0.025}(20-2-2), |t| \geq t_{0.025}(16)$$

$$|t| \geq 2.12$$

5) Calculation:

$$t = \frac{0.5 \sqrt{16}}{\sqrt{1 - (0.51)^2}} \Rightarrow t = 2.37$$

Conclusion: Since the computed value $t = 2.37$
falls in C.R so we can reject H_0 .

How to find least value of r ? $\alpha = 0.05$

- 1) Find calculated value. $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$
- 2) Find table value.

$$|t| \geq t_{\alpha/2}(n-2)$$

- 3) For significant value compare calculated and table value.

$$n = 18, \quad \alpha = 0.05$$

$$\text{calculated value: } t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \Rightarrow t = \frac{r\sqrt{16}}{\sqrt{1-r^2}}$$

$$t = \frac{48}{\sqrt{1-r^2}}$$

Table value:

$$|t| \geq t_{\alpha/2}(n-2) \Rightarrow |t| \geq t_{0.025}(16), \quad |t| \geq 2.12$$

For a significant value of r

$$\frac{48}{\sqrt{1-r^2}} \geq 2.12 \Rightarrow \text{Taking square on b.sides.}$$

$$16r^2 \geq 4.494 - 4.494r^2 \quad 16r^2 \geq (4.494)(1-r^2)$$

$$16r^2 \geq 4.494 - 4.494r^2$$

$$16r^2 + 4.494r^2 \geq 4.494 \Rightarrow 20.494r^2 \geq 4.494$$

$$r^2 \geq \frac{4.494}{20.494} \Rightarrow r^2 \geq 0.219$$

$$r \geq 0.47$$

$$r = 0.47, \quad \text{critical value} = 2.12$$

$$t > 2.12$$

$$\frac{r\sqrt{n-2}}{\sqrt{1-r^2}} > 2.12 = \frac{0.47\sqrt{n-2}}{\sqrt{1-0.47^2}} > 2.12$$

$$\sqrt{n-2} > \frac{(2.12)\sqrt{1-0.47^2}}{0.47}$$